A KRR Mechanism in In-Context Transformers

Condensed Lecture Notes

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1 The Question

Pointwise agreement with KRR is weak: a transformer may satisfy $F(y) \approx Ty$ at one label vector without implementing the KRR label-response rule nearby. We therefore seek a layered certificate.

1. **Fixed-geometry operator.** Freeze

   $$G = (X_{\text{ctx}}, X_{\text{tgt}})$$

   and vary only y. Then KRR is the linear operator $T = K_t A^{-1}$.

2. **Behavior.** Test the local label response

   $$\text{does the transformer's local response } D_\varepsilon F(y)[u] \text{ match } Tu?$$

   on task-distributed directions $u \sim \mathcal{N}(0, A)$, not arbitrary directions.

3. **Subspace sufficiency.** Can we extract from the context residual stream H_ℓ^{ctx} a basis Q such that ridge still works when its dual search is restricted to $\text{span}(Q)$? With $Q^\top A Q = I$, the restricted dual solution is $\alpha_Q(y) = QQ^\top y$. Since predictions are read through K_t, the natural induced operator is

   $$T_Q = K_t Q Q^\top.$$

   Ask whether this restriction commits little prediction-relevant error, $\rho_G(T_Q)$ small, and whether the transformer's response follows it, $E_\varepsilon(F, T_Q)$ small.

4. **Layerwise availability.** Compute the intrinsic prediction-risk rank r_T from the finite KRR task, then ask where response spans C_ℓ first contain a prefix $Q_{\ell,k}$ with $k \approx r_T$ and $\rho_G(T_{Q_{\ell,k}}) \leq 0.05$.

5. **On-manifold use.** Convert the certified dual subspace into a label-space projection

   $$y_Q = AQQ^\top y, \qquad y_\perp = y - y_Q.$$

   Since $Ty_Q = K_t Q Q^\top y = T_Q y$, the test is direct: the model should behave normally on y_Q, while $y_\perp$ should not contain the KRR-relevant prediction signal.

The claim is local: in successful regimes, the transformer realizes the task-distributed KRR label-response operator, explained by low-dimensional activation subspaces whose required dimension is computed from the finite KRR task before seeing activations.

2 Fixed Geometry Makes Ridge Regression an Operator

Fix one episode:

$$X_{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times d_x}, \qquad X_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d_x}.$$

The fixed geometry is

$$G = (X_{\text{ctx}}, X_{\text{tgt}}).$$

During operator tests, only

$$y = (y_1, \dots, y_{n_{\text{ctx}}}) \in \mathbb{R}^{n_{\text{ctx}}}$$

varies. Define

$$K = X_{\text{ctx}} X_{\text{ctx}}^\top, \qquad K_t = X_{\text{tgt}} X_{\text{ctx}}^\top, \qquad A = K + \sigma^2 I.$$

Linear-kernel ridge regression gives

$$\alpha^\star = A^{-1} y, \qquad y^\star = K_t \alpha^\star = K_t A^{-1} y.$$

Thus fixed-geometry ridge regression is the linear map

$$T_G := K_t A^{-1}, \qquad T_G : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}.$$

The transformer defines

$$F_\theta^G : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}, \qquad F_\theta^G(y) = \widehat{y}_\theta.$$

When G is clear, write T and F. The core test is not only $F(y) \approx Ty$, but whether the local label-response operator of F matches T.

3 Context-Label Directions, Local Operators, and Additivity

A context-label direction is

$$u \in \mathbb{R}^{n_{\text{ctx}}},$$

and perturbing labels in direction u means

$$y + \varepsilon u = (y_1 + \varepsilon u_1, \dots, y_{n_{\text{ctx}}} + \varepsilon u_{n_{\text{ctx}}}).$$

Pointwise agreement checks

$$F(y) \approx Ty.$$

Since ridge is linear,

$$J_T(y) = T \quad \text{for every } y.$$

The local version of "same label-to-prediction rule" is

$$J_F(y) u \approx T u,$$

or, for small δ,

$$F(y + \delta) \approx F(y) + T\delta.$$

Estimate the Jacobian action by the central finite difference

$$D_\varepsilon F(y)[u] := \frac{F(y + \varepsilon u) - F(y - \varepsilon u)}{2\varepsilon}.$$

The behavioral operator test is

$$D_\varepsilon F(y)[u] \approx Tu.$$

This is tangent-level agreement only. Matching value and derivative at one point does not imply global equality.

The main probe law is task-label:

$$u \sim \mathcal{N}(0, A).$$

Indeed, under the matched linear task

$$y = X_{\mathrm{ctx}}\beta + \eta, \qquad \beta \sim \mathcal{N}(0, I), \qquad \eta \sim \mathcal{N}(0, \sigma^2 I),$$

conditioning on G gives

$$y \mid G \sim \mathcal{N}(0, X_{\mathrm{ctx}} X_{\mathrm{ctx}}^\top + \sigma^2 I) = \mathcal{N}(0, A),$$

so

$$\mathrm{Cov}(y \mid G) = A.$$

Equivalently,

$$u = A^{1/2} z, \qquad z \sim \mathcal{N}(0, I).$$

The isotropic comparison law is

$$u \sim \mathcal{N}(0, I).$$

The claim is task-local because task probes are emphasized by the data-generating distribution; isotropic probes test arbitrary context-label directions.

For held-out probes $U = \{u_1, \ldots, u_m\}$ and a fixed linear operator $S : \mathbb{R}^{n_{\mathrm{ctx}}} \to \mathbb{R}^{n_{\mathrm{tgt}}}$, define

$$E_\varepsilon(F, S; U) := \left(\frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - S u_j\|_2^2}{\sum_{j=1}^m \|T u_j\|_2^2 + \eta_{\mathrm{num}}} \right)^{1/2}.$$

When $S = T$, this measures local KRR response error, normalized by KRR response energy.

Ridge is additive:

$$T(u + v) = Tu + Tv.$$

A nonlinear F need not satisfy

$$D_\varepsilon F(y)[u + v] \approx D_\varepsilon F(y)[u] + D_\varepsilon F(y)[v].$$

For paired directions $V = \{(u_j, v_j)\}_{j=1}^m$, define

$$C_j = F(y + \varepsilon u_j + \varepsilon v_j) - F(y + \varepsilon u_j - \varepsilon v_j) - F(y - \varepsilon u_j + \varepsilon v_j) + F(y - \varepsilon u_j - \varepsilon v_j),$$

and

$$A_\varepsilon(F; V) := \left(\frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j + \varepsilon v_j)\|_2^2 + \eta_{\mathrm{num}}} \right)^{1/2}.$$

For any affine label map, $C_j = 0$. This diagnostic checks whether a local linear operator comparison is meaningful at the chosen ε and probe law.

4 Activation Subspaces and Prediction-Risk Rank

Behavioral certificates are

$$F(y) \approx Ty, \qquad D_\varepsilon F(y)[u] \approx Tu.$$

The subspace question is: can we extract from the context residual stream a basis Q such that, if the KRR dual solution is forced to lie in $\mathrm{span}(Q)$, the induced predictor still behaves like full KRR in prediction-relevant directions? This makes the reduced operator a constrained KRR object, not an arbitrary fitted readout.

Ridge uses context-position dual variables:

$$\alpha^\star(y) = A^{-1}y \in \mathbb{R}^{n_{\mathrm{ctx}}}, \qquad Ty = K_t\alpha^\star(y).$$

At layer ℓ, the context residual stream is

$$H_\ell^{\mathrm{ctx}} \in \mathbb{R}^{n_{\mathrm{ctx}} \times D}.$$

Each column $H_\ell^{\mathrm{ctx}}[:,k]$ is a context-position pattern in $\mathbb{R}^{n_{\mathrm{ctx}}}$. Hence linear combinations of residual-stream columns can define candidate subspaces for the dual variable α.

Let

$$Q = [q_1, \dots, q_r] \in \mathbb{R}^{n_{\mathrm{ctx}} \times r}, \qquad \mathrm{span}(Q) = \{Qc : c \in \mathbb{R}^r\} \subseteq \mathbb{R}^{n_{\mathrm{ctx}}}.$$

Use an A-orthonormal basis:

$$Q^\top A Q = I.$$

This is the ridge dual geometry, since the quadratic energy is $\frac{1}{2}\alpha^\top A\alpha$.

Full ridge solves

$$\alpha^\star(y) = \arg\min_{\alpha \in \mathbb{R}^{n_{\mathrm{ctx}}}} \left\{ \frac{1}{2}\alpha^\top A\alpha - \alpha^\top y \right\},$$

whose first-order condition gives $\alpha^\star = A^{-1}y$. Restrict to $\alpha = Qc$:

$$\frac{1}{2}c^\top Q^\top AQc - c^\top Q^\top y = \frac{1}{2}c^\top c - c^\top Q^\top y.$$

Then

$$c^\star = Q^\top y, \qquad \alpha_Q(y) = QQ^\top y.$$

Without A-orthonormality,

$$\alpha_Q(y) = Q(Q^\top AQ)^{-1}Q^\top y.$$

Predictions use K_t, so once the restricted dual solution is $\alpha_Q(y) = QQ^\top y$, the corresponding induced operator is forced to be

$$T_Q := K_t QQ^\top.$$

This is natural because it is exactly the full KRR prediction rule $K_t\alpha$, with only the allowed dual search space changed. It is not a free probe: the readout is KRR-constrained.

Prediction relevance is task-weighted. Conditional on G,

$$y \mid G \sim \mathcal{N}(0, A), \qquad u \sim \mathcal{N}(0, A).$$

The reference target is the latent noise-free vector

$$f_{\mathrm{tgt}} = \big(f(x_1^{\mathrm{tgt}}), \dots, f(x_{n_{\mathrm{tgt}}}^{\mathrm{tgt}})\big) \in \mathbb{R}^{n_{\mathrm{tgt}}},$$

with $K_{tt} = X_{\rm tgt} X_{\rm tgt}^\top$ in the linear case. Under the matched GP/KRR model,

$$\begin{bmatrix} y \\ f_{\rm tgt} \end{bmatrix} \Bigm| G \sim \mathcal{N}\left(0, \begin{bmatrix} A & K_t^\top \\ K_t & K_{tt} \end{bmatrix}\right).$$

Gaussian conditioning gives

$$\mathbb{E}[f_{\rm tgt} \mid y, G] = K_t A^{-1} y = Ty$$

and

$$\mathrm{Cov}(f_{\rm tgt} \mid y, G) = K_{tt} - K_t A^{-1} K_t^\top.$$

We need a scale for approximation error. Exact KRR is the posterior mean, but it is not perfect: even after observing y, the latent target values still have posterior uncertainty. That unavoidable uncertainty is the irreducible KRR posterior prediction risk:

$$R_{\rm KRR}(G) := \mathbb{E}\|f_{\rm tgt} - Ty\|_2^2 = \mathrm{tr}\Big(K_{tt} - K_t A^{-1} K_t^\top\Big).$$

For any linear approximation S, decompose

$$f_{\rm tgt} - Sy = \big(f_{\rm tgt} - Ty\big) + \big(T - S\big)y.$$

The posterior residual is orthogonal to every function of y. Indeed,

$$\mathbb{E}[f_{\rm tgt} - Ty \mid y, G] = \mathbb{E}[f_{\rm tgt} \mid y, G] - Ty = 0,$$

so for any measurable $g(y)$,

$$\mathbb{E}\left[\langle f_{\rm tgt} - Ty, g(y)\rangle \mid G\right] = \mathbb{E}\left[\langle \mathbb{E}[f_{\rm tgt} - Ty \mid y, G], g(y)\rangle \mid G\right] = 0.$$

Taking $g(y) = (T - S)y$ kills the cross-term, hence

$$\mathbb{E}\|f_{\rm tgt} - Sy\|_2^2 = R_{\rm KRR}(G) + \mathbb{E}_{y\sim\mathcal{N}(0,A)}\|(T - S)y\|_2^2.$$

Define the excess-risk ratio

$$\rho_G(S) := \frac{\mathbb{E}_{y\sim\mathcal{N}(0,A)}\|(T - S)y\|_2^2}{R_{\rm KRR}(G) + \eta_{\rm num}}.$$

Here S is any candidate label-to-target linear operator: it may be an arbitrary low-rank approximation, or later the activation-constrained operator T_Q. The ratio says exactly how much posterior prediction risk is added by replacing exact KRR T with S:

$$\mathbb{E}\|f_{\rm tgt} - Sy\|_2^2 = R_{\rm KRR}(G) + \rho_G(S)\big(R_{\rm KRR}(G) + \eta_{\rm num}\big).$$

Ignoring the tiny numerical stabilizer, $\rho_G(S) = 0.05$ means “five percent above KRR risk”.

The prediction-risk rank asks a different, intrinsic question: if we are allowed to choose the best rank-r linear replacement for T, how large must r be before $\rho_G(S)$ is at most the chosen tolerance? To answer this, rewrite the same excess-risk term in whitened task-label coordinates.

Whiten task-label directions:

$$y = A^{1/2} z, \qquad z \sim \mathcal{N}(0, I).$$

Then

$$Ty = K_t A^{-1} A^{1/2} z = K_t A^{-1/2} z.$$

Define the whitened KRR response operator

$$C_G := K_t A^{-1/2}.$$

For any candidate S, define its whitened response

$$B_S := S A^{1/2}.$$

Then

$$\mathbb{E}_{y\sim\mathcal{N}(0,A)}\|(T-S)y\|_2^2 = \mathbb{E}_{z\sim\mathcal{N}(0,I)}\|(C_G - B_S)z\|_2^2 = \|C_G - B_S\|_F^2.$$

This is the important coordinate change. The approximation error is not $\|C_G - S\|_F^2$. The operator S acts on unwhitened labels y, while C_G acts on whitened coordinates z. Therefore S must be moved into the same whitened coordinates before comparison:

$$S \quad \longmapsto \quad B_S = S A^{1/2}.$$

Thus S has not disappeared; it appears as $SA^{1/2}$. The rank problem is the problem of approximating C_G by $B_S = SA^{1/2}$, and then mapping back to label coordinates by $S = B_S A^{-1/2}$.

Let

$$C_G = U \operatorname{diag}(s_i) V^\top, \qquad s_1 \geq s_2 \geq \cdots \geq 0.$$

Since z is isotropic,

$$\mathbb{E}\|C_G z\|_2^2 = \sum_i s_i^2.$$

By Eckart–Young, the best rank-r whitened response keeps the first r singular modes. Write

$$C_{G,r} := U_{:,1:r} \operatorname{diag}(s_1,\ldots,s_r) V_{:,1:r}^\top, \qquad S_r := C_{G,r} A^{-1/2}.$$

Then S_r is the best rank-r label-to-target operator for the task-weighted excess-risk metric. It is chosen precisely so that its whitened version is the truncated KRR response:

$$S_r A^{1/2} = C_{G,r}.$$

Therefore it leaves

$$\sum_{i>r} s_i^2.$$

The tail is not the risk of C_G by itself. It is the excess-risk term for the particular oracle choice $S = S_r$:

$$\mathbb{E}_{y\sim\mathcal{N}(0,A)}\|(T-S_r)y\|_2^2 = \|C_G - S_r A^{1/2}\|_F^2 = \|C_G - C_{G,r}\|_F^2 = \sum_{i>r} s_i^2.$$

Its total prediction risk is

$$R_{\mathrm{KRR}}(G) + \sum_{i>r} s_i^2.$$

Definition 1 (Prediction-risk rank)**.** *For fixed geometry G and tolerance $\alpha > 0$, define*

$$r_\alpha(G) := \min\left\{ r : \sum_{i>r} s_i^2 \leq \alpha R_{\mathrm{KRR}}(G) \right\}.$$

In the experiments,

$$r_T := r_{5\%},$$

so r_T is the smallest rank whose best reduced KRR response has prediction risk at most five percent above exact KRR:

$$\mathbb{E}\|f_{\text{tgt}} - S_r y\|_2^2 \le 1.05\,\mathbb{E}\|f_{\text{tgt}} - Ty\|_2^2 = 1.05\,R_{\text{KRR}}(G),$$

where S_r is the optimal rank-r reduced KRR response in whitened task-label coordinates. With $\eta_{\text{num}} = 0$, the same statement for any reduced operator S is

$$\rho_G(S) \le 0.05 \quad \Longleftrightarrow \quad \mathbb{E}\|f_{\text{tgt}} - Sy\|_2^2 \le 1.05\,\mathbb{E}\|f_{\text{tgt}} - Ty\|_2^2.$$

With the stabilizer present, this equivalence is understood up to the negligible η_{num} correction. This is not the algebraic rank of T. It is the number of task-visible KRR modes needed for prediction equivalence at the chosen tolerance. A particular activation subspace Q of dimension r_T still has to align with those modes; success requires $\rho_G(T_Q) \le 0.05$.

The activation-subspace test is therefore representation-dependent. The intrinsic rank r_T uses the oracle operator S_r, which is built directly from the singular modes of C_G. The actual certificate uses $S = T_Q$, where Q is extracted from hidden states and the readout is forced to be the KRR readout $K_t Q$. Equal dimension alone does not imply equal risk.

For an activation basis Q, set

$$C_G = K_t A^{-1/2}, \qquad W = A^{1/2} Q, \qquad P_Q = WW^\top.$$

For $u = A^{1/2} z$,

$$Tu = C_G z, \qquad T_Q u = K_t Q Q^\top A^{1/2} z = C_G P_Q z.$$

Thus

$$\rho_G(T_Q) = \frac{\|C_G(I - P_Q)\|_F^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

If W spans the top r right singular directions of C_G, then

$$\rho_G(T_Q) = \frac{\sum_{i>r} s_i^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

Hence $\dim \operatorname{span}(Q) \ge r_T$ is necessary but not sufficient; alignment with leading task-label modes is also required.

Remark 1 (Reading the numbers)**.** $\rho_G(S) = 0.05$ *means that using S instead of exact KRR adds five percent of the KRR posterior prediction risk. The finite-difference errors $E_\varepsilon(F, T)$ and $E_\varepsilon(F, T_Q)$ are separate operator-response diagnostics.*

The mechanism checks are

$$E_\varepsilon(F, T; U) \text{ small}, \qquad \rho_G(T_Q) \text{ small}, \qquad E_\varepsilon(F, T_Q; U) \text{ small}.$$

The decomposition

$$D_\varepsilon F(y)[u] - Tu = \big(D_\varepsilon F(y)[u] - T_Q u\big) + \big(T_Q u - Tu\big)$$

separates model-subspace mismatch from reduced-KRR approximation error.

To extract an A-orthonormal subspace from any context-pattern matrix $M \in \mathbb{R}^{n_{\text{ctx}} \times p}$, compute

$$A^{1/2}M = U\Sigma V^\top,$$

keep selected left singular directions, and map back:

$$Q(M) := A^{-1/2}U_{\text{kept}}.$$

Then $Q(M)^\top AQ(M) = I$. The raw final-state extractor is

$$M_{\text{raw}} = H_L^{\text{ctx}}, \qquad Q_{\text{raw}} := Q(M_{\text{raw}}).$$

The response-only extractor uses

$$Z_L(u) = \frac{H_L^{\text{ctx}}(y + \varepsilon u) - H_L^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}, \qquad M_{\text{resp}} = [Z_L(u_1), \ldots, Z_L(u_m)],$$

and

$$Q_{\text{resp}} := Q(M_{\text{resp}}).$$

After extraction, Q is frozen. The certificate is a fixed activation-derived T_Q that is prediction-risk sufficient and matches held-out local responses.

5 Layerwise Availability Construction

The layerwise diagnostic asks when a KRR-sufficient response subspace becomes available. At layer ℓ,

$$Z_\ell(u) := \frac{H_\ell^{\text{ctx}}(y + \varepsilon u) - H_\ell^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stack build probes:

$$M_\ell = [Z_\ell(u_1), \ldots, Z_\ell(u_m)].$$

An A-weighted SVD gives

$$A^{1/2}M_\ell = U_\ell \Sigma_\ell V_\ell^\top, \qquad q_i = A^{-1/2}U_{\ell,i}.$$

Write the singular values as

$$\sigma_{\ell,1} \geq \sigma_{\ell,2} \geq \cdots .$$

There are two separate steps. Conceptually, we can keep all singular directions of $A^{1/2}M_\ell$ and test all prefixes. In exact arithmetic this means keeping all directions with $\sigma_{\ell,i} > 0$. In finite precision, near-zero singular directions are unstable and mostly encode numerical noise, so the implementation uses only a mild numerical guard:

$$d_\ell = \min\left\{r_{\max}, \#\{i : \sigma_{\ell,i} \geq \tau_{\text{sv}}\sigma_{\ell,1}\}\right\},$$

with $r_{\max}$ the candidate-rank cap. Taking $\tau_{\text{sv}} = 0$ would correspond to trying all nonzero directions. Then the activation-derived candidate response span is

$$C_\ell = [q_1, \ldots, q_{d_\ell}], \qquad C_\ell^\top AC_\ell = I.$$

Thus C_ℓ is the context-position span made visible by label motion at layer ℓ, up to finite-precision cleanup. This truncation is not the KRR sufficiency criterion; it only prevents near-null numerical directions from entering the candidate activation span.

The principled step comes next. Inside C_ℓ, order directions by KRR usefulness. If C_ℓ is A-orthonormal, set

$$W_\ell = A^{1/2} C_\ell, \qquad Z = K_t A^{-1/2}.$$

The eigenvectors of

$$W_\ell^\top Z^\top Z W_\ell$$

ordered from largest to smallest define nested prefixes

$$Q_{\ell,0} \subset Q_{\ell,1} \subset \cdots \subset C_\ell.$$

This produces the best KRR-aligned prefixes available inside the activation-derived span. It is an availability test, not a claim about the transformer's internal basis choice. For each prefix,

$$T_{Q_{\ell,k}} = K_t Q_{\ell,k} Q_{\ell,k}^\top$$

and

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}}) A^{1/2}\|_F^2}{R_{\mathrm{KRR}} + \eta_{\mathrm{num}}}.$$

The intrinsic prediction-risk rank is $r_T(G; 0.05)$. The representation-dependent rank to five percent is

$$k_\ell^{5\%} := \min\{k : \rho_G(T_{Q_{\ell,k}}) \le 0.05\}.$$

If $k_\ell^{5\%} \approx r_T$, the layer contains an almost optimally aligned KRR subspace. If $k_\ell^{5\%} \gg r_T$, relevant information is present but poorly aligned. If no k exists, the candidate span is insufficient.

6 Experiment 1: Final-State Operator Certificate

Experiment 1 certifies the endpoint: local KRR behavior plus a compact final-state context subspace whose induced T_Q explains that behavior. All checkpoint names below refer to `checkpoints/final/` when archived.

$E(F, S)$ is the held-out finite-difference operator error. $\rho_G(S)$ is exact excess prediction risk. Pointwise prediction errors use the unperturbed y, e.g. $\|F(y) - Ty\| / \|Ty\|$.

The checkpoint is `linear_baseline_dx5_L8.pt`, copied from `checkpoints/model_L8.pt`: an 8-layer linear-regression ICL transformer with $d_x = 5$, $D = 128$, 4 heads, no layer norm. The evaluation uses

$$K = X_{\mathrm{ctx}} X_{\mathrm{ctx}}^\top, \qquad K_t = X_{\mathrm{tgt}} X_{\mathrm{ctx}}^\top,$$

$$n_{\mathrm{ctx}} = 47, \qquad n_{\mathrm{tgt}} = 3, \qquad \sigma^2 = 0.1.$$

Episode sampling:

$$c \sim \mathrm{Unif}[1, 10], \qquad x_i \sim \mathcal{N}(0, cI_5), \qquad \beta \sim \mathcal{N}(0, I_5),$$

$$y_i = x_i^\top \beta + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

The larger-probe rerun uses 4 episodes, seed 123, 64 build probes, 128 held-out probes, 16 additivity pairs, 64 interpolation probes, and

$$\varepsilon \in \{3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}, 3 \cdot 10^{-1}, 10^0\}.$$

Table 1: Experiment 1 endpoint certificate, mean over 4 episodes. $E(F, S)$ is held-out finite-difference operator error on task-label probes. $\rho_G(S)$ is exact prediction-risk excess relative to KRR.

Check	Result	Reading
Model vs. exact KRR	$E_{\text{task}}(F, T) = 0.01284$; pointwise discrepancy 0.01045	Local response is KRR-like on task-label perturbations.
Raw final basis	$k_{\rho \le 5\%} = 6$, $\rho_G(T_Q) = 0.00439$, $E_{\text{task}}(F, T_Q) = 0.01252$	Final residual stream contains a sufficient KRR source subspace.
Response final basis	$k_{\rho \le 5\%} = 6$, $\rho_G(T_Q) = 0.00698$, $E_{\text{task}}(F, T_Q) = 0.01279$	The sufficient subspace is visible in label-induced hidden-state motion.
Same-span readout control	raw: $E_{\text{task}}(F, T_{\text{actLR}}) = 0.00644$, $\rho_G(T_{\text{actLR}}) = 0.1260$	A free readout fits F better but is less KRR-like.
Matched random subspaces	$\rho_G(T_Q) \approx 40.6$ at raw rank, ≈ 33.3 at response rank	Rank alone does not explain sufficiency.
Isotropic label probes	$E_{\text{iso}}(F, T) \approx 0.292$	The certificate is task-local, not global ridge equality.

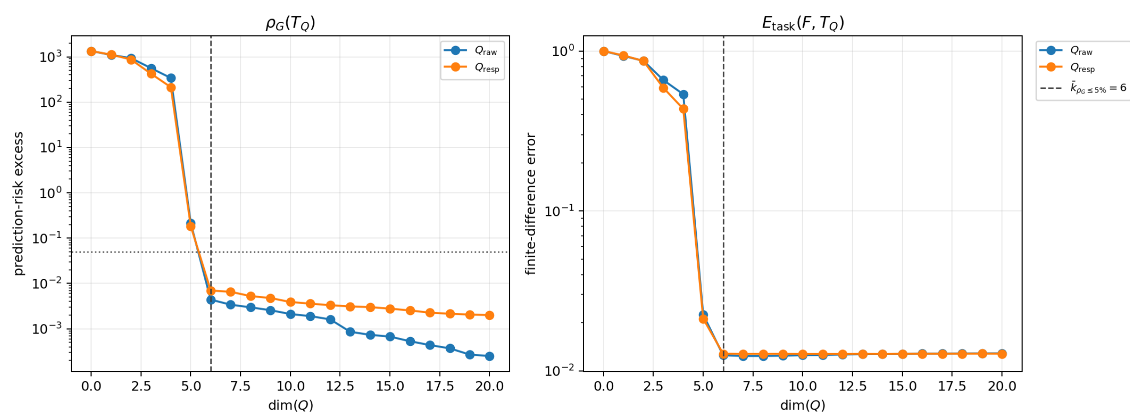

Figure 1: Experiment 1 rank-prefix curves. Left: $\rho_G(T_Q)$, with the five-percent tolerance. Right: $E_{\text{task}}(F, T_Q)$. The dashed line marks $k_{\rho \le 5\%} = 6$.

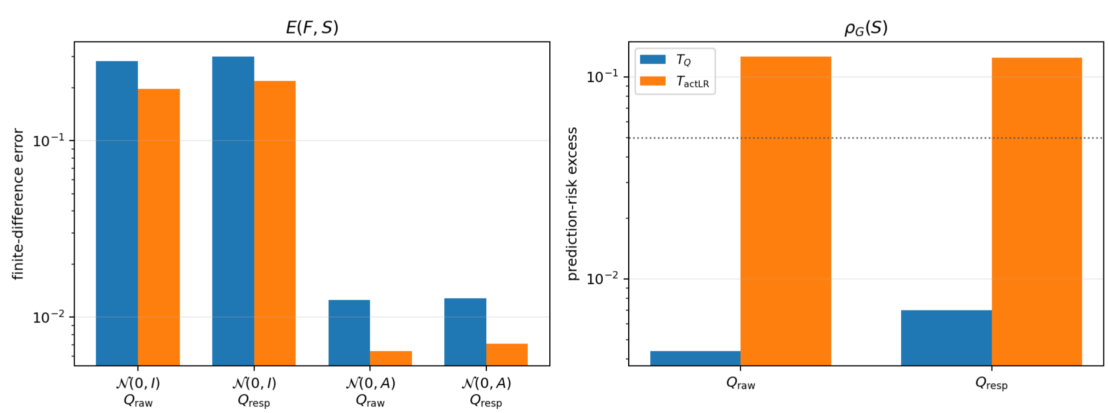

Figure 2: Experiment 1 same-span readout control. actLR uses the same activation coordinates $Q^\top u$ but fits a free least-squares readout; T_Q uses the KRR-constrained readout $K_t Q$.

The endpoint certificate below reports the $\varepsilon = 3 \cdot 10^{-3}$ run. It uses $\tau_{\rm sv} = 10^{-3}$, selection tolerance 0.05, and rank curves up to 20. Raw and response bases are A-weighted SVD prefixes; select the first rank satisfying

$$\rho_G(T_Q) \leq 0.05.$$

In the reported run, both raw and response bases reach this at rank 6.

For probe-law interpolation,

$$u_\lambda = \sqrt{1-\lambda}\, u_{\rm task} + \sqrt{\lambda}\, u_{\rm iso}, \qquad \lambda \in [0, 1],$$

where $u_{\rm task} \sim \mathcal{N}(0, A)$ and $u_{\rm iso} \sim \mathcal{N}(0, I)$. $\lambda = 0$ is task-label probing; $\lambda = 1$ is isotropic probing. A fresh matched response basis Q_λ is built for each λ.

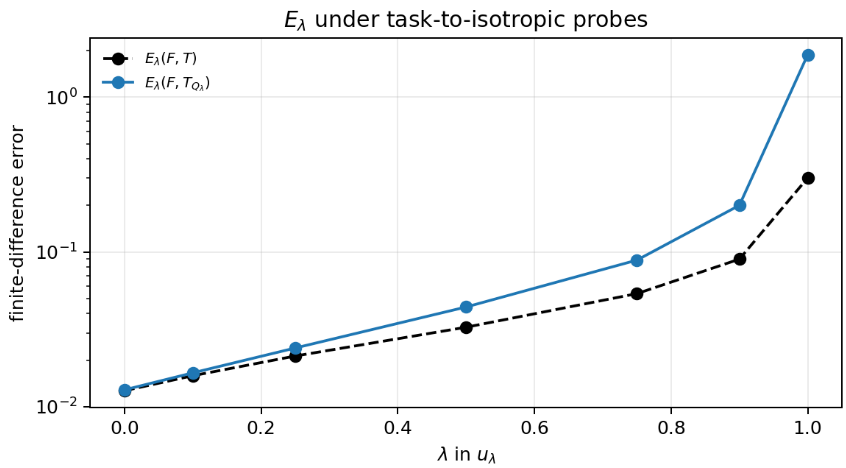

Figure 3: Experiment 1 task-to-isotropic interpolation. KRR-like local behavior degrades as perturbations leave the task-distributed directions.

At $\varepsilon = 3 \cdot 10^{-3}$, the normalized additivity defect is

$$1.41 \times 10^{-4}$$

on task probes and

$$1.79 \times 10^{-3}$$

on isotropic probes. Thus the finite-difference operator comparison is locally meaningful.

The epsilon sweep checks that this is not a fragile finite-difference scale choice:

ε	$E_{\rm task}(F, T)$	$E_{\rm task}(F, T_{Q,\rm raw})$	$E_{\rm task}(F, T_{Q,\rm resp})$	$A_\varepsilon(F)$ task
$3 \cdot 10^{-3}$	0.01284	0.01252	0.01279	$1.41 \cdot 10^{-4}$
10^{-2}	0.01284	0.01252	0.01280	$2.77 \cdot 10^{-4}$
$3 \cdot 10^{-2}$	0.01284	0.01252	0.01279	$8.13 \cdot 10^{-4}$
10^{-1}	0.01284	0.01251	0.01279	$2.57 \cdot 10^{-3}$
$3 \cdot 10^{-1}$	0.01276	0.01243	0.01269	$6.55 \cdot 10^{-3}$
10^0	0.01237	0.01207	0.01230	$1.43 \cdot 10^{-2}$

The operator estimate is stable through $3 \cdot 10^{-1}$ and still reasonable at 10^0. The additivity defect is smallest near $3 \cdot 10^{-3}$ and rises at 10^{-1}–10^0, so $\varepsilon = 3 \cdot 10^{-3}$–$10^{-2}$ is the clean finite-difference window for the certificate.

7 Experiment 2: Prediction-Risk Rank and Layerwise Availability

Experiment 2 separates the intrinsic KRR rank from representation-dependent availability. The intrinsic rank r_T is computed from

$$C_G = K_t A^{-1/2}$$

before seeing activations. The layerwise rank $k_\ell^{5\%}$ asks how many directions from the response span $C_\ell = \mathrm{span}(D_y H_\ell^{\mathrm{ctx}})$ are needed for

$$\rho_G \le 0.05.$$

Prefixes are evaluated by

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}})A^{1/2}\|_F^2}{R_{\mathrm{KRR}} + \eta_{\mathrm{num}}}.$$

In tables, "cand. dim." is $\dim C_\ell$, "rank to 5%" is $k_\ell^{5\%}$, "best excess" is $\min_k \rho_G(T_{Q_{\ell,k}})$, and "reach" is the fraction of episodes where some prefix reaches the five-percent target.

All Experiment 2 runs use $\varepsilon = 10^{-3}$, $\alpha = 0.05$, $\sigma^2 = 0.1$, 16 build task probes, 16 held-out task probes, 8 layers, $D = 128$, 4 heads, and candidate basis kinds response/raw/combined; the main tables report the response basis unless stated otherwise. There is no artificial rank cap in the reported diagnostics: after numerical SVD cleanup by τ_{sv}, every prefix $k = 0, \dots, \dim C_\ell$ is evaluated. For each episode,

$$A = K + \sigma^2 I, \qquad T = K_t A^{-1}.$$

The flat linear family uses

$$c \sim \mathrm{Unif}[1, 10], \qquad x_i \sim \mathcal{N}(0, cI_{d_x}), \qquad \beta \sim \mathcal{N}(0, I_{d_x}),$$

$$y_i^{\mathrm{ctx}} = x_i^{\mathrm{ctx}\top}\beta + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma^2), \qquad y_i^{\mathrm{tgt}} = x_i^{\mathrm{tgt}\top}\beta,$$

and evaluates with the linear kernel

$$K = X_{\mathrm{ctx}} X_{\mathrm{ctx}}^\top, \qquad K_t = X_{\mathrm{tgt}} X_{\mathrm{ctx}}^\top, \qquad K_{tt} = X_{\mathrm{tgt}} X_{\mathrm{tgt}}^\top.$$

The $d_x = 40$ checkpoint keeps the same linear evaluation kernel but samples a random covariance $\Sigma = U\,\mathrm{diag}(\lambda) U^\top$, then

$$x_i \sim \mathcal{N}(0, \Sigma), \qquad \tau(\Sigma) = \sum_{j=1}^{40} \frac{\lambda_j}{\lambda_j + \sigma^2} \le 16.$$

The native training window is

$$\log_{10} \lambda_1 \in [-2, 2], \qquad n_{\mathrm{tgt}} = 8, \qquad n_{\mathrm{ctx}} = \mathrm{round}(\gamma\tau(\Sigma)), \qquad \gamma \in \{4, 8, 12\}.$$

The spectrum is either a smooth log-decay or a step spectrum, with $\tau(\Sigma)$ sampled in the feasible part of $[10^{-3}, 16]$. The $d_x = 40$ diagnostic below reconstructs the native validation bank: 30 batches of size 16, seed 122, with each batch sharing one sampled spectrum and one γ. The RBF family uses

$$x_i \sim \mathcal{N}(0, I_5), \qquad \kappa(x, x') = \exp\left(-\frac{\|x - x'\|_2^2}{2\ell^2}\right), \qquad \ell = 3.0,$$

signal variance 1.0, and latent draw

$$f \mid X \sim \mathcal{N}(0, K_{\mathrm{full}} + 10^{-5} I), \qquad y^{\mathrm{ctx}} = f_{\mathrm{ctx}} + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2 I), \qquad y^{\mathrm{tgt}} = f_{\mathrm{tgt}}.$$

For RBF evaluation, K, K_t, K_{tt} are the finite matrices induced by the same κ. Episode counts differ by run:

run	episodes	n_{ctx}	n_{tgt}
small linear $d_x \in \{3, 5, 8, 10, 15\}$	16 each	47	16
main RBF 128×128	8	128	128
$d_x = 40$ native validation bank	480	3–187	8
older fixed-$\ell = 3$ RBF $n_{\text{tgt}} \in \{16, 64, 128\}$	16 each	47	n_{tgt}

Seeds are 42 except for the $d_x = 40$ runs, which use seed 122. The RBF runs use lengthscale 3.0 and signal variance 1.0. The main RBF run uses $\tau_{\text{sv}} = 10^{-4}$; the other Experiment 2 runs use $\tau_{\text{sv}} = 10^{-3}$.

The linear sweep uses `linear_sweep_dx3.pt`, `linear_sweep_dx5.pt`, `linear_sweep_dx8.pt`, `linear_sweep_dx10.pt`, and `linear_sweep_dx15.pt`. All are 8-layer, $D = 128$, 4-head linear checkpoints. For $d_x = 3, 5, 8, 10, 15$, with

$$n_{\text{ctx}} = 47, \qquad n_{\text{tgt}} = 16,$$

the diagnostic recovers $r_T \approx d_x$ up to the target cap.

Table 2: Experiment 2 linear calibration. "First layer" is the first layer whose response span reaches $\rho_G \leq 0.05$.

Checkpoint	r_T	first layer	final cand. dim.	$k_L^{5\%}$	$E_{\text{task}}(F,T)$
$d_x = 3$	3.0	0	27.5	3.0	0.00617
$d_x = 5$	5.0	0	27.6	5.0	0.00918
$d_x = 8$	8.0	0	40.1	8.0	0.01250
$d_x = 10$	10.0	0	40.4	10.0	0.01433
$d_x = 15$	14.9	2	46.6	14.9	0.01735

The main emergence checkpoint is `rbf_fixed_l3_nctx128_ntgt128_seed42.pt`, trained on a fixed RBF kernel with lengthscale 3.0, $d_x = 5$, $D = 128$, 8 layers, 4 heads, and 8 evaluation episodes,

$$n_{\text{ctx}} = 128, \qquad n_{\text{tgt}} = 128.$$

The model response is locally close to KRR:

$$E_{\text{task}}(F, T) = 0.0306.$$

On these 8 diagnostic episodes,

$$\frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{KRR}}} = 1.03$$

as an average of per-episode ratios. For this finite KRR problem,

$$r_T \approx 24.75.$$

Layer 0 is insufficient. By layer 2, a nearly task-rank subspace is visible; by layers 4 and 8, $k_\ell^{5\%}$ matches r_T within finite-sample noise.

7.1 dx40 Native Bank: Scale-Stratified Availability and Label-Space Use

The $d_x = 40$ checkpoint `dx40_tau16_s122.pt` is a scale-aware 8-layer, $D = 128$, 4-head linear checkpoint trained with

$$n_{\text{tgt}} = 8, \qquad n_{\text{ctx}} = \text{round}(\gamma\tau(\Sigma)), \qquad \gamma \in \{4, 8, 12\},$$

$$\log_{10} \lambda_1 \in [-2, 2], \quad \tau(\Sigma) = \sum_{i=1}^{40} \frac{\lambda_i}{\lambda_i + \sigma^2} \leq 16.$$

Table 3: Experiment 2 main emergence result: selected layers of the RBF 128 × 128 checkpoint. Here $r_T \approx 24.75$.

Layer	cand. dim.	rank to 5%	$\rho_G(T_{Q_{\ell,r_T}})$
0	16	–	1.147
1	51	38.0	0.100
2	84	26.9	0.058
4	107	25.2	0.050
8	119	24.9	0.048

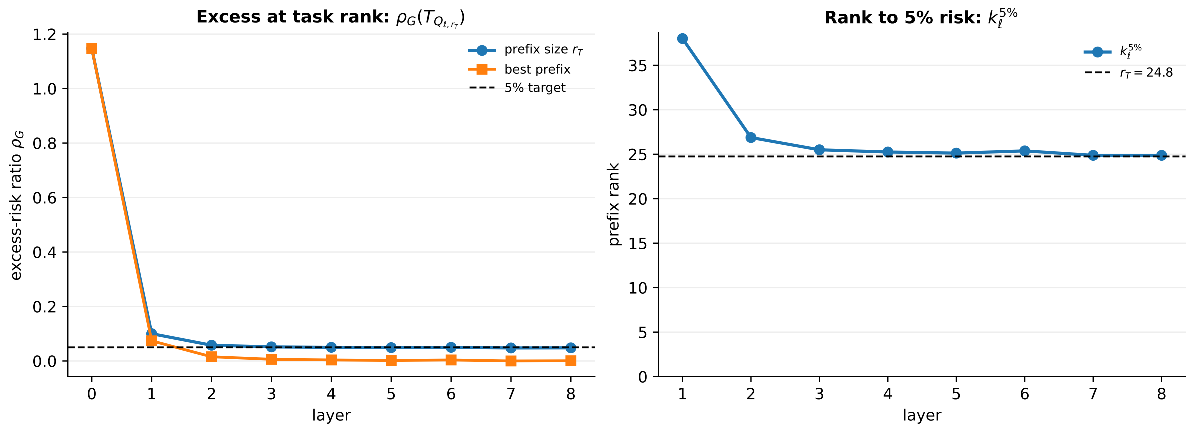

Figure 4: Experiment 2 layerwise emergence in the matched RBF 128 × 128 checkpoint. Left: ρ_G for the r_T-prefix and best prefix. Right: $k_\ell^{5\%}$ against r_T.

The diagnostic uses the native validation bank:

$$n_{\text{tgt}} = 8, \qquad n_{\text{ctx}} = \text{round}(\gamma\tau(\Sigma)), \qquad 480 = 30 \times 16 \text{ episodes}.$$

Across these 480 episodes, n_{ctx} ranges from 3 to 187. The average pointwise ratio is

$$\frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{KRR}}} = 1.638,$$

as an average of per-episode ratios, with median 1.236 and aggregate means

$$\text{MSE}_{\text{model}} = 0.144, \qquad \text{MSE}_{\text{KRR}} = 0.103.$$

Table 4: $d_x = 40$ native validation-bank response-span diagnostic, stratified by scale. "Rank to 5%" is averaged over reaching episodes.

$\log_{10}\lambda_1$	n	reach	r_T	$k_8^{5\%}$	$\rho_G(T_{Q_{8,r_T}})$
$[-2,-1)$	112	1.000	6.42	6.42	0.0167
$[-1,0)$	176	1.000	7.99	7.99	0.0005
$[0,1)$	96	0.927	8.00	8.00	0.0171
$[1,2]$	96	0.188	7.04	3.11	0.2232

The full-bank average says the response basis usually appears:

$$\mathbb{E}[r_T] = 7.44, \qquad \mathbb{E}[k_8^{5\%} \mid \text{reach}] = 7.33, \qquad \Pr(\text{reach at layer 8}) = 395/480 = 0.823.$$

The scale-stratified result is sharper. For $\log_{10}\lambda_1 < 1$, the final response span almost always contains a KRR-sufficient prefix. For $\log_{10}\lambda_1 \in [1, 2]$, the certificate usually fails: only 18/96 episodes reach $\rho_G \le 0.05$. Thus the dx40 model generally forms the relevant basis at low and moderate scale, but high-scale episodes are a distinct failure mode.

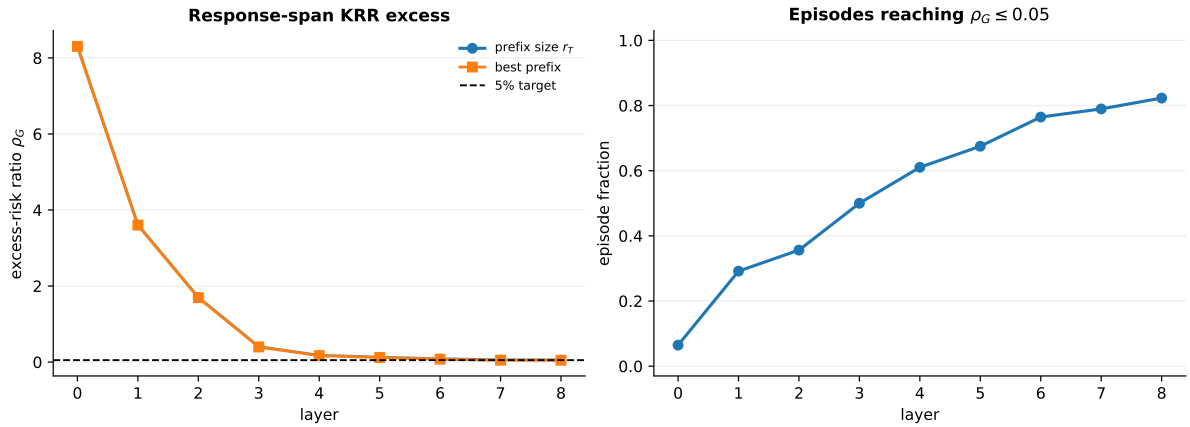

Figure 5: Experiment 2 $d_x = 40$ native validation-bank diagnostic. With all retained prefixes evaluated, the response span increasingly reaches the five-percent target across layers and succeeds on 82.3% of episodes by the final layer.

To test use without leaving the model's normal input manifold, extract the same final response prefix Q, then decompose the context labels by

$$y_Q := AQQ^\top y, \qquad y_\perp := y - y_Q.$$

This is the natural label-space version of the reduced dual solution, because

$$Ty_Q = K_t A^{-1} AQQ^\top y = K_t QQ^\top y = T_Q y.$$

Thus y_Q asks whether the model can still solve the task when the label stream contains only the certified KRR-relevant component, while $y_\perp$ asks whether the complementary labels contain the KRR prediction signal. The model is run normally from scratch on each label stream.

Table 5: $d_x = 40$ native-bank label-space projection by scale. Entries are medians over episodes in the scale bin. The y_Q and $y_\perp$ columns are MSE/KRR ratios after feeding the model the projected labels $AQQ^\top y$ and the complementary labels $y - AQQ^\top y$.

$\log_{10}\lambda_1$	r_T	k	$\rho_G(T_Q)$	base MSE/KRR	y_Q MSE/KRR	$y_\perp$ MSE/KRR
$[-2, -1)$	7	7	$1.32 \cdot 10^{-2}$	1.07	1.06	2.90
$[-1, 0)$	8	8	$2.23 \cdot 10^{-26}$	1.16	1.03	39.9
$[0, 1)$	8	8	$3.03 \cdot 10^{-4}$	1.53	1.15	156
$[1, 2]$	8	15	0.197	2.13	1.59	828

On the whole native bank,

$$\text{median}\,\frac{\mathrm{MSE}(F(y_Q))}{\mathrm{MSE}_{\mathrm{KRR}}} = 1.104, \qquad \text{median}\,\frac{\mathrm{MSE}(F(y_\perp))}{\mathrm{MSE}_{\mathrm{KRR}}} = 40.28.$$

The low- and moderate-scale bins show the intended pattern: the projected labels y_Q preserve KRR-level prediction, while $y_\perp$ loses the target-relevant signal. At high scale, $\rho_G(T_Q)$ is often not certified, and both the base model and the label projection degrade.

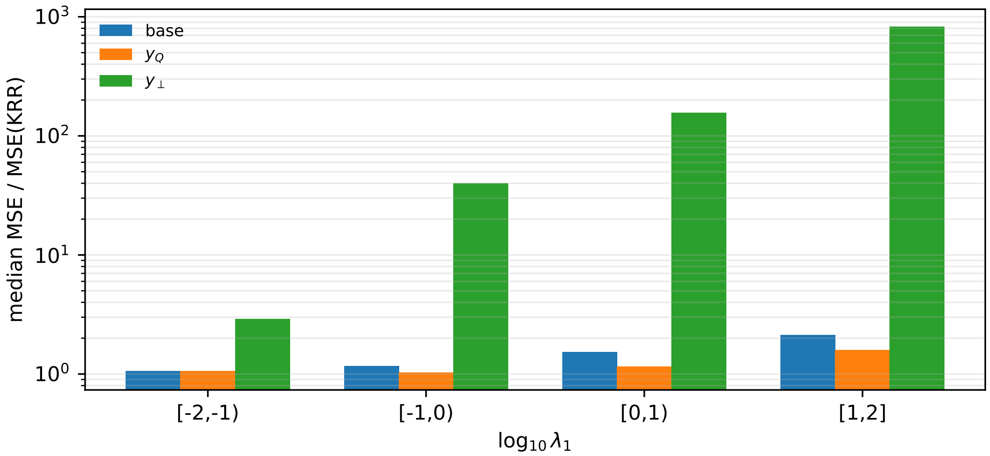

Figure 6: $d_x = 40$ native-bank label-space projection. Bars report median MSE/KRR by scale bin for the original labels y, the certified component $y_Q = AQQ^\top y$, and the complement $y_\perp = y - y_Q$.

Table 6: Older fixed-$\ell = 3$ RBF checkpoint, final-layer response diagnostic.

n_{tgt}	r_T	$k_L^{5\%}$	$\rho_G(T_{Q_{L,r_T}})$	$E_{\text{task}}(F,T)$
16	10.6	11.1	0.049	0.067
64	17.1	17.8	0.053	0.074
128	18.3	19.3	0.057	0.097

For the older fixed-$\ell = 3$ RBF checkpoint, the final-layer response diagnostic at $n_{\text{ctx}} = 47$ is: Availability can remain near-sufficient while the realized finite-difference operator becomes less clean.

8 Experiment 3: On-Manifold Label-Space Use

Experiments 1–2 show that a KRR-sufficient dual subspace is available. Experiment 3 asks whether that subspace is enough when it is converted back into actual context labels, so the transformer is always run on a valid input sequence.

The checkpoint is `linear_baseline_dx5_L8.pt`. The diagnostic uses flat-spectrum linear episodes with

$$d_x = 5, \qquad n_{\text{ctx}} = 47, \qquad n_{\text{tgt}} = 16, \qquad \sigma^2 = 0.1.$$

Use 32 episodes, seed 31415, raw final-state bases, $\tau_{\text{sv}} = 10^{-3}$, and $r_{\max} = 12$. For each episode, extract

$$Q = [q_1, \ldots, q_k], \qquad Q^\top AQ = I.$$

The average extracted size is $k = 11.8$, while the intrinsic KRR risk rank is

$$\mathbb{E}[r_T] = 5.$$

The induced reduced operator is highly sufficient:

$$\mathbb{E}\,\rho_G(T_Q) = 1.55 \cdot 10^{-3}.$$

Now define

$$y_Q := AQQ^\top y, \qquad y_\perp := y - y_Q.$$

Since

$$Ty_Q = K_t A^{-1} AQQ^\top y = K_t QQ^\top y = T_Q y,$$

feeding y_Q is the on-manifold version of evaluating the certified reduced KRR operator. Feeding $y_\perp$ tests the complementary label component. No hidden residual state is edited.

Table 7: Experiment 3 label-space projection. Ratios are episode MSE divided by the original KRR MSE. The KRR-input column reports the same label projection passed through KRR itself.

Input labels	mean MSE/KRR	median MSE/KRR	median KRR-input/KRR	median drift from $F(y)$
y	1.619	1.085	1.000	0
$y_Q = AQQ^\top y$	1.602	1.090	0.9999	$2.91 \cdot 10^{-4}$
$y_\perp = y - y_Q$	$5.27 \cdot 10^3$	$1.77 \cdot 10^3$	$1.77 \cdot 10^3$	1.000

Thus the projected label stream y_Q preserves the model's prediction and the KRR oracle. The complement $y_\perp$ is not a useful label stream: KRR itself fails on it by the same scale as the model. This is the clean use test: the certified dual subspace carries the target-relevant label information without requiring off-manifold hidden-state edits.

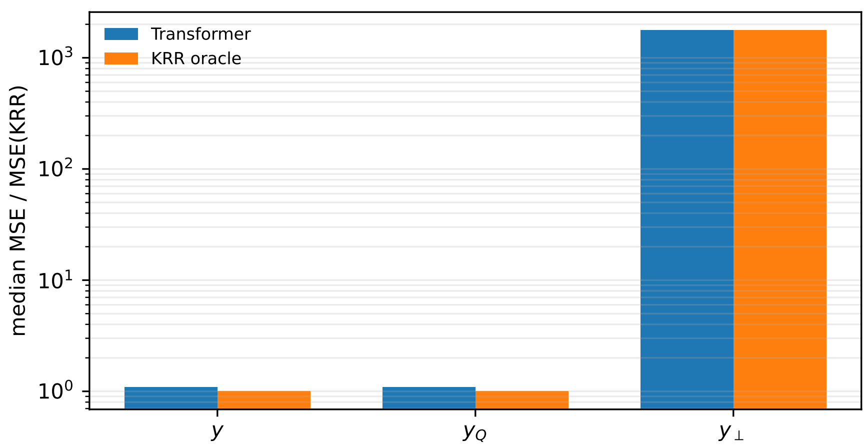

Figure 7: Experiment 3 on-manifold label-space projection. The transformer and KRR oracle both remain near the original KRR error on y_Q, while both fail on $y_\perp$.

9 Experiment 4: Repair by Label-Space Projection

Experiment 4 is a conditional repair test. First choose a setting where the model usually builds a KRR-sufficient response basis. Then look only at the rare episodes where the model output is bad. Finally test whether the on-manifold label projection $y_Q = AQQ^\top y$ repairs those bad outputs.

Use the $d_x = 40$ checkpoint `dx40_tau16_s122.pt` in the target-rich diagnostic setting

$$n_{\text{ctx}} = 47, \qquad n_{\text{tgt}} = 40, \qquad \log_{10} \lambda_1 \in [-2, 1].$$

This is not the native validation distribution. It is a controlled setting in which the response prefix usually exists, so the label-space projection has a chance to be principled.

For each episode, form the final-layer response candidate span

$$C_8 = \operatorname{span}(D_y H_8^{\mathrm{ctx}}),$$

order it by KRR-targeted energy, and choose the first prefix satisfying

$$Q = Q_{8,k}, \qquad k = \min\{j : \rho_G(T_{Q_{8,j}}) \le 0.05\}.$$

If no such prefix exists, record the episode as not certified.

Sample a 200-episode bank with seed 20260506. Before conditioning on outliers, the bank has the following distribution:

quantity	mean	median	q_{25}–q_{75}	tail / range
$\mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{KRR}}$	84.96	1.14	1.03–1.37	$q_{90} = 1.67,\ q_{99} = 4.19,\ \max = 1.67 \cdot 10^4$
r_T	23.29	25	19.75–28	1–32
k	23.80	25	20–29	1–41
$\rho_G(T_Q)$	0.0444	0.0441	0.0403–0.0471	$\Pr(\rho_G(T_Q) \le 0.05) = 194/200$

Thus the bank is mostly non-pathological and basis-present: the median MSE ratio is 1.14, the certificate holds on 194/200 = 97% of episodes, and the typical KRR rank and extracted prefix size are both around 23–25. The large arithmetic mean MSE ratio is caused by one catastrophic outlier, bank id 35; excluding only this largest outlier gives a non-outlier mean near 1.28.

After this bank-level scan, select episodes where the unmodified model is bad:

$$\frac{\mathrm{MSE}_{\mathrm{model}}}{\mathrm{MSE}_{\mathrm{KRR}}} \ge 5.$$

Only two episodes pass. For each selected episode, run the transformer normally on

$$y, \qquad y_Q = AQQ^\top y, \qquad y_\perp = y - y_Q.$$

The model is never rolled from an edited hidden state. This matters because the certificate $\rho_G(T_Q)$ is a statement about the KRR dual subspace, and y_Q is the label stream whose ordinary KRR prediction is exactly $T_Q y$.

Table 8: Experiment 4 label-space repair. The bank has 200 target-rich episodes; two satisfy $\mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{KRR}} \ge 5$. Rows report averages over the two selected episodes.

Input labels	mean MSE/KRR	median MSE/KRR	Reading
y	$8.37 \cdot 10^3$	$8.37 \cdot 10^3$	pathological output
$y_Q = AQQ^\top y$	1.079	1.079	near-KRR repair
$y_\perp = y - y_Q$	$6.00 \cdot 10^3$	$6.00 \cdot 10^3$	complementary labels fail

The two selected episodes are:

bank id	r_T	k	$\rho_G(T_Q)$	base MSE/KRR	y_Q MSE/KRR	$y_\perp$ MSE/KRR
35	2	3	0.0482	$1.67 \cdot 10^4$	1.169	$1.20 \cdot 10^4$
184	1	1	0.0371	6.833	0.989	7.393

Both outliers have $\rho_G(T_Q) \le 0.05$, so the KRR-sufficient response basis is present before conditioning on bad model outputs. Projecting the labels onto that basis is enough to bring the model back to KRR-level error on both outliers. The complement does not repair: it carries little of the target-relevant KRR signal.

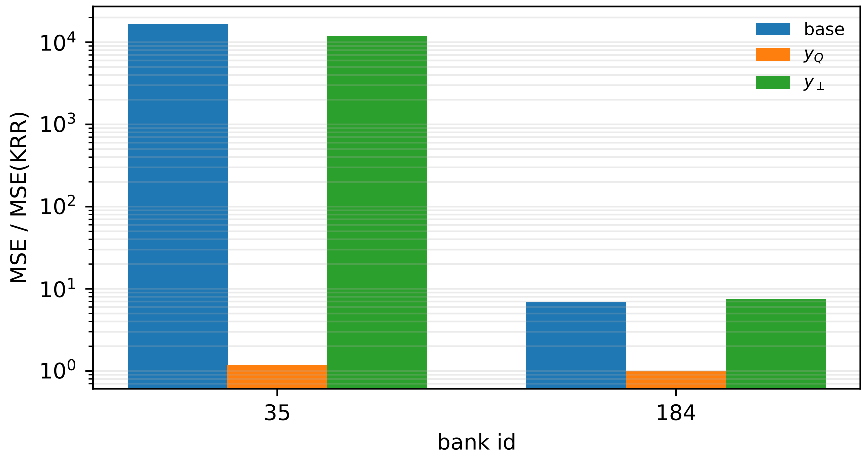

Figure 8: Experiment 4 target-rich outliers. Feeding $y_Q = AQQ^\top y$ repairs the two bad episodes; feeding $y_\perp = y - y_Q$ does not.

10 Logical Chain

1. Freeze the episode geometry $G = (X_{\text{ctx}}, X_{\text{tgt}})$.
2. Under fixed G, ridge regression becomes $T = K_t A^{-1}$.
3. Treat the transformer as a label map $F : \mathbb{R}^{n_{\text{ctx}}} \to \mathbb{R}^{n_{\text{tgt}}}$.
4. Estimate the local response $D_\varepsilon F(y)[u]$.
5. Test behavior with $E_\varepsilon(F, T)$ on task-label probes.
6. Compute r_T from $C_G = K_t A^{-1/2}$ and R_{KRR}.
7. Extract activation-derived context-side candidate spans C_ℓ.
8. Order each candidate span by KRR-targeted energy and form prefixes $Q_{\ell,k}$.
9. Construct $T_Q = K_t Q Q^\top$.
10. Test subspace sufficiency with $\rho_G(T_Q)$.
11. Test model-subspace alignment with $E_\varepsilon(F, T_Q)$.
12. Convert certified Q into label projections $y_Q = AQQ^\top y$ and $y_\perp = y - y_Q$.
13. Run the model normally on y_Q and $y_\perp$ to test on-manifold use and repair.

Mechanism story: behavioral similarity, KRR-structured activation certificate, risk-rank sufficiency, layerwise availability, and on-manifold label-space use.